COURSE TITLE : BUILDING SERVICES
COURSE CODE : 6181
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Electrical wiring systems	15
2	Plumbing and water supply	15
3	Sanitation and Fire fighting systems	15
4	Vertical transportation, HVAC systems and Acoustics.	15
	Total	**60**

Course Objective :

Sl.	Sub	Student will be able to
I	1	Understand the services in a building
	2	Analyse various systems of services applicable in a building
	3	Understand electrical lighting systems in a building Understand working of water distribution systems Understand the working of an efficient and economic sanitary system in a building Select suitable fire-fighting measures for a building Understand parts and working of lifts and escalators Identify and select suitable air-conditioner for a building Analyse acoustical requirements of a building

Upon completion of the course the student should be able to:

Module I ELECTRICAL WIRING SYSTEMS

1.1. Understand the services in a building

1.1.1. List out different services needed in a building

1.2. Understand electrical wiring for lighting in a building

1.2.1.Define wiring system
1.2.2.Differentiate between properties of consumer and supplier
1.2.3.Categorise systems of distribution of electrical energy
1.2.4.Classify methods of wiring and systems of wiring
1.2.5.Classify conductor material used in cables, insulating materials, wires and cables electrical fittings and Accessories
1.2.6. .State function and positioning of equipments like main switch board, meter board, distribution board, fuse used in wiring system of a residence

1.3. Understand electrical shock. it's effects and protective measures to be taken

1.3.1. List out causes of electric shock and precautions to be taken against electric shock
1.3.2.Identify protective devices used in wiring

1.4.Understand earthing

1.4.1.State purpose of earthing and specifications regarding earthing
1.4.2. List out factors affecting earth resistance
1.4.3.Describe methods of earthing

1.5.Identify suitable method of lightning protection of building

1.5.1. State need for iightning protection in building
1.5.2.Identify method of lightning protection in a building

1.6.Identify and draw circuite diagrams

1.6.1.Classify lighting circuites
16.2. Illustrate circuite diagrams of one lamp controlled by one switch and two lamp controlled By two switches
1.6.3..Describe two-way switching
1.6.4.Illustrate bedroom lighting circuite
1.6.5. Describe wiring installation in small residence
1.6.6.State function and positioning of equipments and various outlets used in wiring system of a residence
1.6.7.Know the symbols used in electrical diagrams
1.6.3..Describe two-way switching
1.6.4.Illustrate bedroom lighting circuite

1.6.5. Describe wiring installation in small residence
1.6.6.State function and positioning of equipments and various outlets used in wiring system of a residence
1.6.7.Know the symbols used in electrical diagrams

Module II : PLUMBING AND WATER SUPPLY

2.1.Understand working of public water distribution systems

2.1.1.Define water supply system
2.1.2.list out requirements of water supply system
2.1.3.State quality of water for supply
2.1.4.Differentiate between potable water,wholesome water,polluted water,infected water
2.1.5.State stages of conveyance of water
2.1.6. Categorise water distribution system
2.1.7. Describe method of distribution and layout of distribution

2.2. Select pipe material, pipe size ,pipe joints ,pipe fittings, types of valves needed and water meter for water supply system

2.2.1.Classify pipes according to materials
2.2.2.Classify pipe joints
2.2.3.Classify pipe fittings
2.2.4.Classify valves
2.2.5.Classify pumps
2.2.5.Classify storage tanks
2.2.6.Draw service connections in a building

2.3. Understand hot water supply in a building

2.3.1.Illustrate service connection and layout of hot water supply system in a building
2.3.2. Select suitable geysers for hot water supply in a building

2.4. Understand rain water harvesting

2.4.1.State the need for rain water harvesting system for a building
2.4.2. Illustrate rain water harvesting system for a building

Module IV: FIRE PROTECTION,VERTICAL TRANSPORTATION AND ACOUSTICS

4.1.Identify fire-fighting requirements of a building

4.1.1.State the necessity of fire fighting services in high rise buildings
4.1.2.Catagorise fire
4.1.3.Catagorise building according to fire load
4.1.4.State causes and effects of fire
4.1.5.State important considerations in fire protection and to limit fire spread

4.2. Select suitable means of fire protection in a building

4.2.1. Describe fire-resistant construction of wall and column, floor and roof, wall openings and fire escape elements like stair, entrances and corridors
4.2.2. Explain working of various fire protection systems
4.2.3. Identify essential features and components of a strong room

4.3. Understand parts and working of lifts and escalators

4.3.1. List out different means for vertical transportation
4.3.2. Identify different parts of a lift
4.3.3. Classify lifts
4.3.4. List out structural provisions for lift installation
4.3.5. Select suitable lift and lift arrangement for a building
4.3.6. Identify different parts of an escalator
4.3.7. Select suitable arrangement of escalators for a building

4.4. Analyse acoustical requirements of a building

4.4.1. Define acoustics
4.4.2. State characteristics of audible sound and behaviour of sound

4.5. Solve acoustical defects in a hall or room

4.5.1. Identify acoustical defects and solve it
4.5.2. Classify acoustical materials
4.5.3. Select suitable acoustical materials for different purposes
4.5.4. Explain how acoustical treatments are given in open air theatres, Cinema theatres, radio broadcasting studios, concert halls, multi-purpose halls, public lecture halls and class lecture rooms

CONTENT DETAILS

MODULE-I- ELECTRICAL WIRING SYSTEMS -

Building services-Introduction-necessity-Electrical wiring-introduction-wiring system-consumer's terminal-property of supplier- systems of distribution of electrical energy-distribution board system-tree system-methods of wiring-joint box system-loop in system-systems of wiring-cleat wiring-wooden casing and capping wiring-CTS or PVC or TRS wiring-lead sheathed or metal sheathed wiring-conduite wiring-surface or open-recessed or concealed-wires and cables-difference between wires and cables-parts of a cable-conductor-insulation-protective covering-conductor materials used in cables-copper –aluminium-insulating materials-properties-types-rubber-VIR-impregnated paper-PVC-silk and cotton-types of wires and cables used in internal wiring-according to conductor used-copper conductor cables-aluminium conductor cables-classification of cables and wires according to number of cores-single core cable-twin core cable-three core cable--classification of cables and wires according to voltage grading-250/440 v-650/1100v --classification of cables and wires according to inslation used-VIR-TRS-Lead sheathed-PVC-weather proof-flexible-XLPE-definition and positioning of equipments like meter board,main switch board,distribution board,fuse-Lighting accessories and fittings-switches-one way-two way-two way centr off-double pole-push button-table lamp-bed switches-tumbler switch-flush switches-ceiling rose-socket outlet-plugs-lamp holder-function-classification-BC-ES-GES-batten-pendant-angle-slanting-bracket-water tight-miniature screw type-electrical shock-causes-effects of electrical current on human body-precautions-protective devices-fuses-types of fuses-supply main-consumer main-sub circuite-point-fuse units-round type-kit kat type-cartridge-HRC-MCB-ELCB-RCCB-earthing-purpose-specification regarding earthing-factors affecting earth resistance-methods of earthing-strip-rod-pipe-plate earthing-lightning-different methods of lightning protection-conventional symbols used in electrical installations for residential buildings-lighting sub-circuites-circuite and sub circuite-types-series-parallel-circuite diagrams of-one lamp controlled from one point-socket outlet controlled by one pole switch-two lamps controlled by individual switches-two way switching- wiring installations in small residential building

MODULE –II

PLUMBING AND WATER SUPPLY

Water supply-water supply system-requirements of water supply system-quality of water for public water supply-definition of wholesome water-requirements of wholesome water-definition of potable water, polluted water,contaminated water and infected water-conveyance of water-different stages--distribution system-Definition-systems of distribution-gravity,pumping, gravity and pumping combined-layout of distribution pipes –dead end,grid iron,ring,radial-pipe appurtenances -types of pipes-cast iron pipes,steel pipes,R.C.C pipes, pre-stressed concrete pipes,AC pipes,copper,brass, lead ,wrought iron,PVC,hume steel pipes,polyethelene pipes--pipe joints-types-spigot and socket joint,flanged joint,expansion joint,flexible joint,collar joint,joint for AC pipes,scewed and socketed joint -definition-valves-types-stop valve,check valve,air releif valve,pressure relief valve,scour valve,fire hydrant,plug valves,water meter-pipe fittings- objectives-bow,tee,reducer,lateral,cross,cap,coupling,union,nipple-specifications-types of pumps-displacement pump,dynamic pump- service connections in a building-storage tanks-function-types-underground and overhead tanks-layout of watersupply system in a house-hot and cold water supply system in a building- rain water harvesting system-objectives

MODULE-III

SANITARY ENGINEERING AND HVAC

Sanitary engineering-definition-terminology-house drainage-parts of a drainage system-traps-function-requirements of a good trap-classification of traps according to shape-S,P and Q- classification of traps according to uses-floor trap,gully trap,intercepting trap-grease and oil trap-systems of plumbing-single stack partially ventilated, two pipe,one pipe,single stack-sanitary fixtures-water closet,urinal,sink,wash basin,bath tubs,shower fittings,cubicles,flushing cisterns-manholes-function-location-construction-drop manhole-function-septic tank-working-design and construction details-soak pit-function-construction details-layout of house drainage-ventilation-necessity-functional requirements-systems of ventilation-natural-wind effect and stack effect-mechanical ventilation systems-exhaust ,plenum,balanced and air conditioning-purpose of air conditioning-systems of air conditioning-central,self contained,semi contained,combined-essentials of an air conditioned system-filter,heating,cooling,humidification,dehumidification,air circulation

MODULE IV

FIRE PROTECTION, VERTICAL TRANSPORTATION AND ACOUSTICS

Fire fighting services-classification of fire-class A,class B, class C, class D, class E-Classification of buildings according to fire load-causes and effects of fire-important considerations in fire protection-limiting fire spread-fire resistant construction-fire protection systems-alarm,extinguishers,escape routes-vertical transportation-lifts –types of lifts-traction, hydraulic-passenger lifts, bed lifts,dumb waiters-structural provision for lift installation-lift pit, lift well, machine room ,hoist way-arrangement of lifts-escalators-parts-truss, landing platform, steps, tracks-arrangement -parallel, criss cross ,multiple parallel-acoustics-acoustics of buildings-characteristics of audible sound-frequency, loudness, tone-behaviour of sound-transmission, reflection, absorption-acoustical defects, causes and remedies-echo, reverberation, sound foci and dead spots ,insufficient loudness, noise nuisance-Sabine’s formula-general principles and factors in acoustical design-site selection and planning, volume, shape, treatment of interior surfaces, reverberation, seat ,seating arrangement and audience, sound absorption-acoustical materials-porous absorbents, resonant panel absorbent, cavity resonators, composite type resonators-recommendations for good acoustical design of buildings like open air theatre, cinema theatre, radio broad casting stations, concert halls, class lecture rooms, public lecture halls and multi-purpose theatres

REFERENCE

1. Water supply and Sanitary Engineering-
2. Building construction – Arora and Bindra
3. Building construction --Varghese
4. Building construction---Jha